CSA6303 – THREAT INTELLIGENCE AND NETWORK SECURITY

CO1-A2- Concept Mappting

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats |Introduction to Threat Intelligence Types of Threat Intelligence (Strategic, Tactical, perational)— Threat Intelligence Lifecycle — Applications in IT Industry

Dr.V.Parthipan **Total: 50**

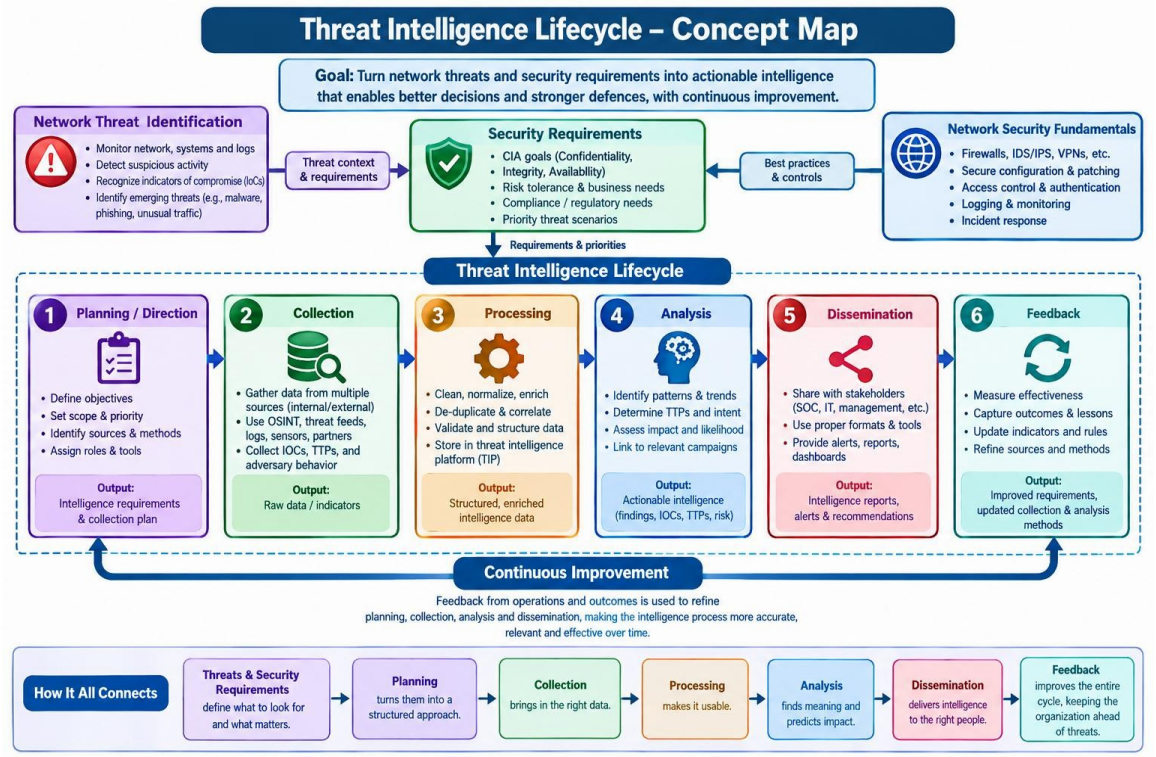
Student Name : C.Amareswar Reddy

Register number:192412470

Code of Conduct:

I **C. Amareswar Reddy (192412470)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : C.Amareswar Reddy



Threat Intelligence Lifecycle – Concept Map
Goal: Turn network threats and security requirements into actionable intelligence that enables better decisions and stronger defences, with continuous improvement.
Network Threat Identification
Monitor network, systems and logs
Detect suspicious activity
Recognize indicators of compromise (IoCs)
Identify emerging threats (e.g., malware, phishing, unusual traffic)
Threat context & requirements
Security Requirements
CIA goals (Confidentiality, Integrity, Availablity)
Risk tolerance & business needs
Compliance / regulatory needs
Priority threat scenarios
Best practices & controls
Network Security Fundamentals
Firewalls, IDS/IPS, VPNs, etc.
Secure configuration & patching
Access control & authentication
Logging & monitoring
Incident response
Requirements & priorities
Threat Intelligence Lifecycle
1 Planning / Direction
Define objectives
Set scope & priority
Identify sources & methods
Assign roles & tools
Output: Intelligence requirements & collection plan
2 Collection
Gather data from multiple sources (internal/external)
Use OSINT, threat feeds, logs, sensors, partners
Collect IOCs, TTPs, and adversary behavior
Output: Raw data / indicators
3 Processing
Clean, normalize, enrich
De-duplicate & correlate
Validate and structure data
Store in threat intelligence platform (TIP)
Output: Structured, enriched intelligence data
4 Analysis
Identify patterns & trends
Determine TTPs and intent
Assess impact and likelihood
Link to relevant campaigns
Output: Actionable intelligence (findings, IOCs, TTPs, risk)
5 Dissemination
Share with stakeholders (SOC, IT, management, etc.)
Use proper formats & tools
Provide alerts, reports, dashboards
Output: Intelligence reports, alerts & recommendations
6 Feedback
Measure effectiveness
Capture outcomes & lessons
Update indicators and rules
Refine sources and methods
Output: Improved requirements, updated collection & analysis methods
Continuous Improvement
Feedback from operations and outcomes is used to refine planning, collection, analysis and dissemination, making the intelligence process more accurate, relevant and effective over time.
How It All Connects
Threats & Security Requirements define what to look for and what matters.
Planning turns them into a structured approach.
Collection brings in the right data.
Processing makes it usable.
Analysis finds meaning and predicts impact.
Dissemination delivers intelligence to the right people.
Feedback improves the entire cycle, keeping the organization ahead of threats.